

Week 5 — AI Book Assistant

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Repos

- **Backend:** <https://github.com/mingjing-zhang/week-05-api>
- **Frontend:** <https://github.com/mingjing-zhang/week-05-frontend>

Both public. API key in `.env`, gitignored.

What's in the chat page

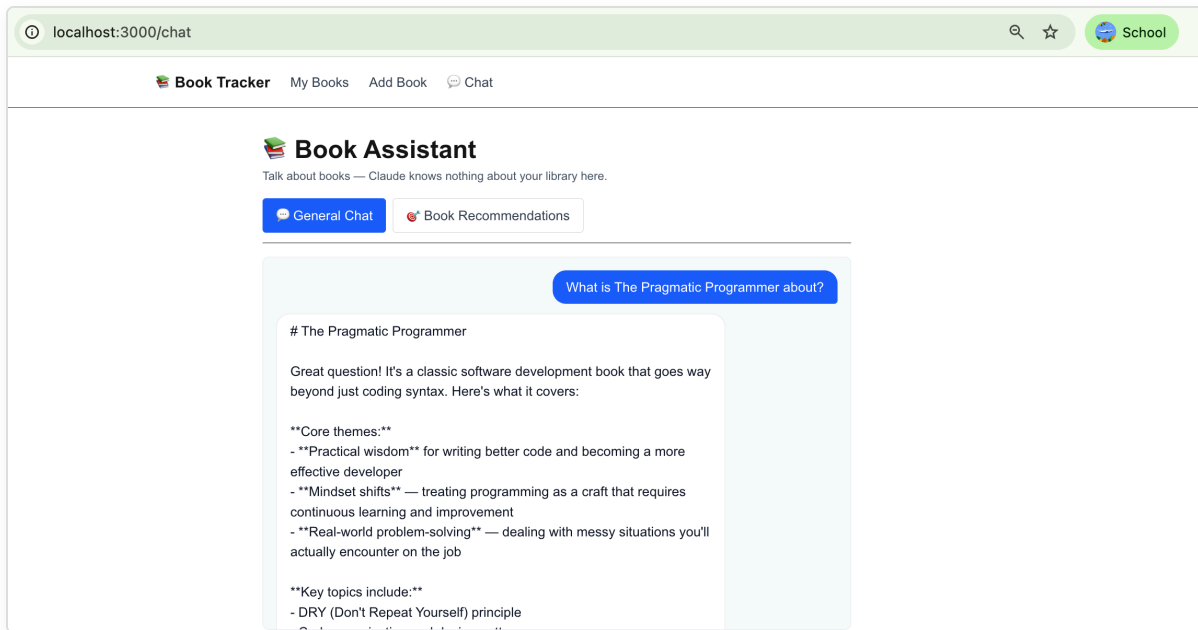
Two modes, toggled at the top:

- **General Chat** → `POST /ai/chat` (no DB read)
- **Book Recommendations** → `POST /ai/recommend` (reads the `Book` table, injects titles into the system prompt)

Right-blue user bubbles, left-white assistant bubbles, conversation history kept in component state and sent on every turn, "Thinking..." indicator, auto-scroll, history clears on mode switch.

Screenshot 1 — General Chat

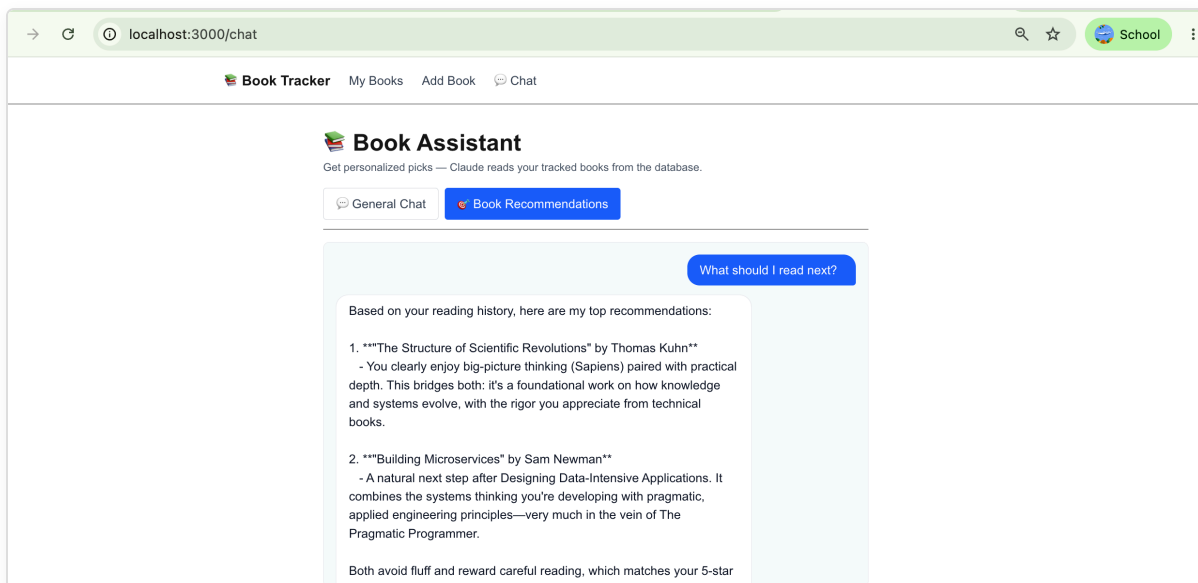
A book question with no DB grounding.



General chat mode

Screenshot 2 — Personalized Recommendation

Recommendation that **explicitly references the user's tracked books** (`Sapiens` , `Designing Data-Intensive Applications`), proving the DB → system-prompt grounding works end to end.



Personalized recommendation

Reflection

Q1. System prompt vs user message — why separate?

The system prompt is developer code; the user message is untrusted input. Mixing them invites prompt injection — the model has no other way to tell which is which. The Anthropic API enforces the separation by giving them different `role` slots. Same shape as SQL parameterization: keep program separate from data.

This is exactly what lets `/ai/recommend` inject the user's book list into the system prompt without ever putting it in the user turn — so the user can't tamper with it from the chat input.

Q2. What happened when you swapped system prompts in Part 4?

Four variants, same user question ("Recommend a book about systems thinking"). Full transcripts in `PROMPT_EXPERIMENTS.md`. Headlines:

- **Baseline:** 201 output tokens, friendly multi-recommendation with a follow-up question.
- **"Marcus" persona** ("opinionated lit professor"): **284 tokens, +41%**. Personas grant permission to editorialize. Cost matters.
- **Strict format constraint** ("3 entries, this template, no preamble"): **165 tokens, the shortest**. Constraints compress output.
- **Topic guardrail** ("ONLY books"): looked identical to baseline on book questions. On *"What's the weather like?"* it cleanly redirected back to books.

The surprise was the guardrail — completely invisible until violated. That's the right shape: silent in the common case, decisive at the edge.

Q3. Where could AI in this app cause harm, and how would you mitigate it?

A confidently-hallucinated book title taken as a purchase decision — money spent on something that doesn't exist. Same shape applies to *"is this book appropriate for my 8-year-old?"*: a fabricated content summary in a parenting context.

Mitigations, in layers:

1. **Ground in real data** (the DB for tracked books; Google Books / OpenLibrary for the rest) rather than model recall.
2. **No irreversible actions.** The model proposes, the user commits. No auto-purchase, no auto-share.
3. **UI transparency.** "AI-generated — verify before action" affordance, link out to a real catalog.
4. **Validate response shape** on the backend for structured outputs, reject hallucinated fields.

The higher the stakes downstream, the more verification belongs between the model and the action.

Q4. If you had infinite credits, what AI feature would you add?

A “**Discuss this book**” mode that grounds the AI in the book’s actual text via RAG.

Concretely: a `book_chunks` table with `pgvector` embeddings; a `POST /books/{id}/ingest_text` endpoint that chunks and embeds full text (Project Gutenberg, partial-content APIs, or user upload); and a `POST /ai/discuss` endpoint that retrieves the top-k passages on each user turn and stuffs them into the system prompt with chunk-index citations. The frontend renders citations as clickable jumps to the source paragraph.

Every other AI feature I might add — summaries, themes, character maps — becomes a downstream query against a grounded source instead of trusting model recall. It also directly closes the Q3 harm: a passage-grounded answer is auditable; a hallucinated summary is not.

There’s a personal angle. My day work is Bitcoin protocol research. The same RAG pattern over BIPs and papers would let me ask “*where exactly did Wuille say that about SIGHASH?*” and get a citable passage. The book tracker is just a friendlier surface for the same pattern.

Notes

- Model: `claude-haiku-4-5` instead of the lab’s `claude-sonnet-4-6` . ~3× cheaper, indistinguishable quality on this workload.
- Total Anthropic spend across endpoint smoke tests, the four Part-4 experiments, and live browser testing: **~\$0.013**.
- `PROMPT_EXPERIMENTS.md` has full transcripts of all four variants plus the off-topic guardrail test.